

MI-03: Resolution

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Repository status: Solved. The complete argument received an independent Codex-agent PASS review against the exact catalog target. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. [Detailed review and scope.](#)

1 MI-03: the contraction constant is $k/4$ for every $k \geq 2$

For a matrix X , write $|X| = (X^*X)^{1/2}$. The target asks for the least dimension-independent constant c_k such that

$$\left| \sum_{j=1}^k A_j \right| \leq c_k I + \sum_{j=1}^k |A_j| \quad (1)$$

for all complex square contractions A_j . Bourin and Lee prove the upper bound $k/4$ and its sharpness for even k , and conjecture sharpness for odd $k > 1$ [1, Corollary 4.4 and Remark 4.5].

Theorem 1.1. *For every integer $k \geq 2$, the optimal constant in (1) is $c_k = k/4$. Sharpness already occurs for complex 2×2 rank-one contractions. It also occurs for Hermitian 3×3 contractions.*

Proof. We first give a short proof of the upper bound. Put

$$R = \left| \sum_{j=1}^k A_j \right|, \quad T = \sum_{j=1}^k |A_j|.$$

For every vector x , the vector Cauchy–Schwarz inequality gives

$$\left\| \sum_j A_j x \right\|^2 \leq k \sum_j \|A_j x\|^2.$$

Consequently,

$$R^2 \leq k \sum_j A_j^* A_j = k \sum_j |A_j|^2 \leq kT,$$

since $0 \leq |A_j| \leq I$. It follows that

$$T + \frac{k}{4}I - R \geq \frac{1}{k}R^2 + \frac{k}{4}I - R = \frac{1}{k} \left(R - \frac{k}{2}I \right)^2 \geq 0.$$

This proves (1) with $c_k = k/4$ without assuming that the summands are Hermitian.

For sharpness, let $e_1 = (1, 0)^T$, let $\omega = e^{2\pi i/k}$, and define

$$v_j = \begin{pmatrix} 1/2 \\ (\sqrt{3}/2)\omega^j \end{pmatrix}, \quad A_j = e_1 v_j^*, \quad 0 \leq j < k.$$

Each v_j is a unit vector. Thus A_j is a rank-one contraction of norm one and

$$|A_j| = v_j v_j^*.$$

The identity $\sum_{j=0}^{k-1} \omega^j = 0$ yields

$$\sum_{j=0}^{k-1} A_j = \frac{k}{2} e_1 e_1^*, \quad \sum_{j=0}^{k-1} |A_j| = \begin{pmatrix} k/4 & 0 \\ 0 & 3k/4 \end{pmatrix}.$$

Hence

$$\left| \sum_{j=0}^{k-1} A_j \right| - \sum_{j=0}^{k-1} |A_j| = \begin{pmatrix} k/4 & 0 \\ 0 & -3k/4 \end{pmatrix}.$$

Taking its quadratic form at e_1 forces $c_k \geq k/4$.

For the Hermitian assertion, work in $\mathbb{C}^3$ and put

$$w_j = \frac{1}{2} e_2 + \frac{\sqrt{3}}{2} \omega^j e_3, \quad H_j = e_1 w_j^* + w_j e_1^*.$$

The orthogonality of e_1 and w_j implies $|H_j| = e_1 e_1^* + w_j w_j^*$ and $\|H_j\|_\infty = 1$. Therefore

$$\sum_j |H_j| = \text{diag}(k, k/4, 3k/4), \quad \left| \sum_j H_j \right| = \text{diag}(k/2, k/2, 0).$$

The quadratic-form difference at e_2 is again $k/4$. □

Remark 1.2 (An explicit positive decomposition). The upper-bound argument can also be checked from the identity

$$\begin{aligned} k \left(T + \frac{k}{4} I - R \right) &= k \sum_j (|A_j| - |A_j|^2) + \sum_{i < j} (A_i - A_j)^* (A_i - A_j) \\ &\quad + \left(R - \frac{k}{2} I \right)^2. \end{aligned}$$

Every term on the right is positive semidefinite. This identity and the root-of-unity construction are exact; numerical verification is not needed for the proof.

References

- [1] Jean-Christophe Bourin and Eun-Young Lee. Diagonal and off-diagonal blocks of positive definite partitioned matrices. arXiv:2307.02034, 2023. Version consulted: v3; Corollary 4.4 and the conjecture following Remark 4.5. URL: <https://arxiv.org/abs/2307.02034>.